



Angeles University Foundation



Training Enrollment System

A Multi-Table PHP PDO Application
Laboratory Activity Documentation

IPT — Integrative Programming and Technology

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Section: [BSIT 3A]

GitHub repository: [<https://github.com/eternalmatrix146-collab/ipt10-lab-01.git>]

1. Project Description

The Training Enrollment System is a web application for an organization that offers training programs. It is built with PHP and PDO and uses a MySQL database with four related tables.

One Administrator uses the system to:

- **Manage courses** — add, edit and delete courses.
- **Manage classes** — each class belongs to a course and has a schedule, an instructor and a number of available slots.
- **Record a student** — save a new student and enroll them into a class in one step.
- **Enroll an existing student** into another class.
- **View enrollments** joined with student, class and course details, with search, status filter and paging.
- **Cancel an enrollment** — the slot goes back to the class.
- **View a report** of active and cancelled enrollments per class.

The system follows three rules:

- **No over-enrollment.** A class with 0 slots shows "No slots available."
- **No duplicate enrollment.** A student cannot have two active enrollments in the same class.
- **Data stays consistent.** Recording a student and cancelling an enrollment each run inside one database transaction. If any step fails, every change is rolled back.

All database work uses PDO with prepared statements.

2. Setup Instructions

Requirements: XAMPP (Apache, MySQL, PHP 7.4 or newer) and a web browser.

1. Copy the project folder into XAMPP so the path is
C:\xampp\htdocs\training_enrollment\
2. Open the XAMPP Control Panel and start Apache and MySQL.
3. Open <http://localhost/phpmyadmin>, click Import, choose the file `sql/schema.sql` and click Import. This creates the `training_db` database with sample data.
4. Copy `config/db.sample.php` and name the copy `config/db.php`. The XAMPP defaults (user root, empty password) already work.
5. Open http://localhost/training_enrollment/
6. Log in with username `admin` and password `admin123`.

3. Database Design

The database has four tables. Foreign keys connect them so the data stays consistent.

Table	What it stores	Relationship
courses	Course code, name and description	One course has many classes
classes	Class code, schedule, instructor and slots still available	Belongs to a course (<code>course_id</code>)
students	Full name, email and phone	One student has many enrollments

enrollments	Which student is in which class, the date and the status (active or cancelled)	Belongs to a student (student_id) and a class (class_id)
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The enrollments table connects students and classes. One student can join many classes, and one class can have many students.

4. Design Patterns

This project uses two design patterns: Singleton and Repository.

4.1 Singleton — Database class

What it is. A Singleton is a class that can only ever have one object.

How it is used. The constructor of the Database class is private, so no page can create it with "new". The only way to get it is Database::getInstance(). The first call creates the connection. Every later call returns the same object.

```
// config/db.php
$db = Database::getInstance($dsn, $user, $pass);
```

Why I used it:

- **One connection per request.** Every class shares the same connection, which is faster than opening a new one each time.
- **Transactions need it.** A transaction belongs to one connection. Recording a student changes three tables. Because all steps use the same connection, a rollback can undo all of them.
- **One place for settings.** The error mode and fetch mode are set once.

4.2 Repository — EnrollmentRepository class

What it is. A Repository is a class that holds all the database code for one topic. Pages ask the repository to do something. They do not write SQL themselves.

How it is used. EnrollmentRepository owns everything about enrollments:

Method	What it does
recordStudent()	Inserts the student, inserts the enrollment and takes one slot, in one transaction
enroll()	Enrolls an existing student and takes one slot, in one transaction
cancel()	Sets the status to cancelled and gives the slot back, in one transaction
allWithDetails()	Lists enrollments joined with student, class and course
search(), countSearch()	The same list with search, status filter and paging
summaryByClass()	The report: active and cancelled enrollments per class

Why I used it:

- **Pages stay simple.** Files in the admin folder only read the form, validate it, call one method and show a message.

- **The transaction rule is written once.** No page can forget a step or do the steps in the wrong order.
- **Easy to change.** If a table changes, only the repository changes.

4.3 How the transaction works

Recording a student runs these steps inside one transaction:

```
beginTransaction()  
  1. Check the class still has slots -> if not: "No slots available."  
  2. INSERT INTO students  
  3. INSERT INTO enrollments  
  4. UPDATE classes SET slots = slots - 1  
commit()
```

If anything fails -> rollBack()
(no student row, no enrollment row, slots unchanged)

Cancelling an enrollment works the same way with two steps: set the status to cancelled, then add one slot back to the class.

5. Screenshot Evidence

Before you start: import sql/schema.sql again so the sample data is fresh. Take the screenshots in order (1 to 9) because later ones depend on earlier ones. Include the browser address bar in every capture. Delete this paragraph and all "How to capture" steps when you are done.

Screenshot 1 — Course Management

How to capture:

1. Open `http://localhost/training_enrollment/admin/courses.php`
2. Capture the page showing the Course List.

Training Enrollment System

Administrator Log In

Username

Password

Log In

[Home](#) | [Courses](#) | [Classes](#) | [Record Student](#) | [Enroll](#) | [Enrollments](#) | [Reports](#) | [Log Out](#)

Training Enrollment System

Welcome, Administrator. Use this system to manage training courses and class schedules, record students, and track enrollments.

- [Courses](#) – add, edit and delete courses
- [Classes](#) – manage class schedules and slots
- [Record Student](#) – record a new student and enroll them
- [Enroll](#) – enroll an existing student into a class
- [Enrollments](#) – view, search and cancel enrollments
- [Reports](#) – enrollment summary per class

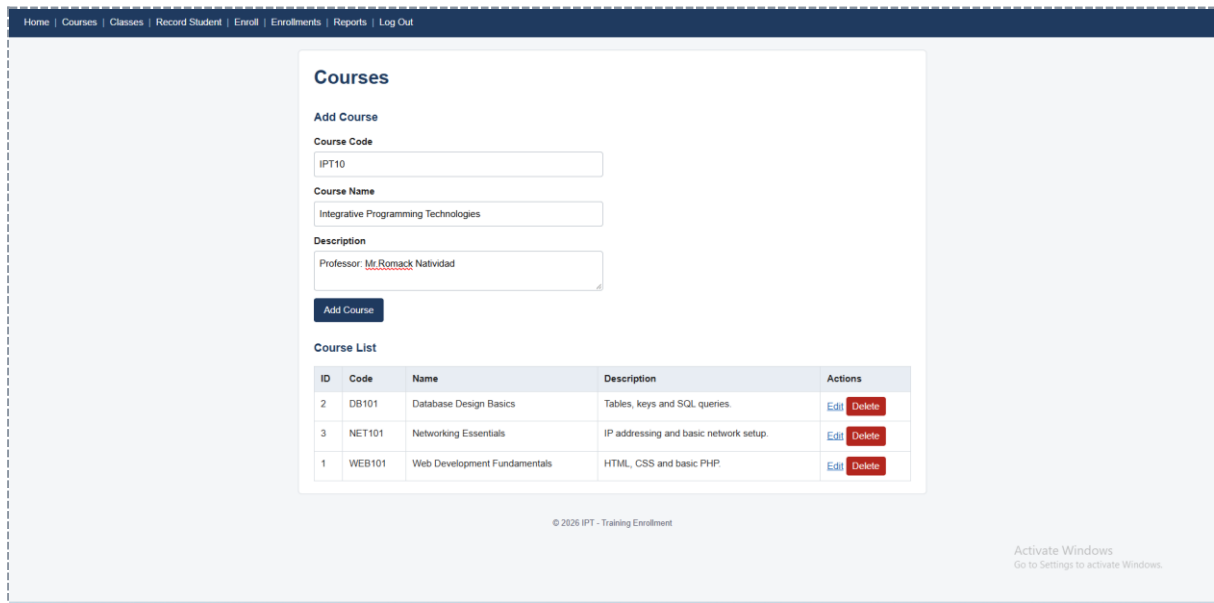
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Activate Windows
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Screenshot 2 — Add / Edit Course Form

How to capture:

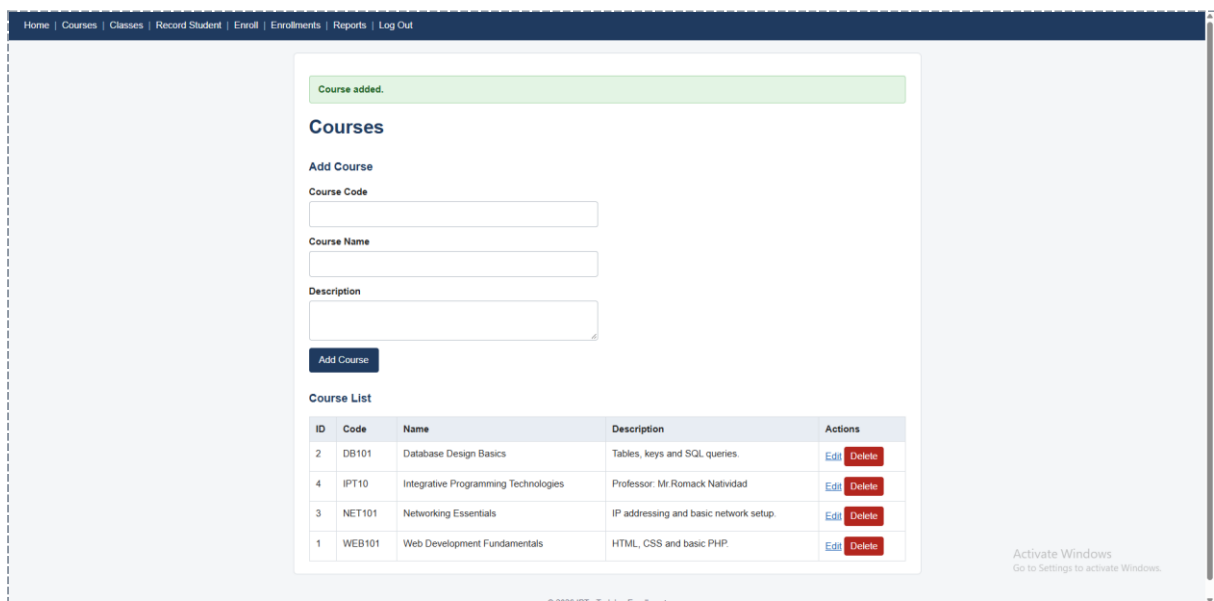
1. On the Courses page, type a new course in the form (for example Code: PY101, Name: Python Basics). Capture it before clicking the button (2a).
2. Click Add Course. Capture the "Course added." message and the new row in the list (2b).



The screenshot shows the 'Courses' management interface. At the top is a navigation bar with links: Home, Courses, Classes, Record Student, Enroll, Enrollments, Reports, and Log Out. The main content area is titled 'Courses' and contains an 'Add Course' form. The form has three input fields: 'Course Code' with the value 'IPT10', 'Course Name' with the value 'Integrative Programming Technologies', and 'Description' with the value 'Professor: Mr. Romack Natividad'. Below the form is a blue 'Add Course' button. Underneath the form is a 'Course List' table. The table has five columns: ID, Code, Name, Description, and Actions. It contains three rows of existing courses. At the bottom right of the page, there is a watermark that says 'Activate Windows Go to Settings to activate Windows.'

ID	Code	Name	Description	Actions
2	DB101	Database Design Basics	Tables, keys and SQL queries.	Edit Delete
3	NET101	Networking Essentials	IP addressing and basic network setup.	Edit Delete
1	WEB101	Web Development Fundamentals	HTML, CSS and basic PHP.	Edit Delete

Figure 1a. Add Course form filled in with a new course.



This screenshot shows the same 'Courses' interface as Figure 1a, but after the new course has been saved. A green success message 'Course added.' is displayed at the top of the form area. The 'Add Course' form fields are now empty. The 'Course List' table now includes the newly added course as a fourth row. The rest of the interface, including the navigation bar and the Windows watermark, remains the same.

ID	Code	Name	Description	Actions
2	DB101	Database Design Basics	Tables, keys and SQL queries.	Edit Delete
4	IPT10	Integrative Programming Technologies	Professor: Mr. Romack Natividad	Edit Delete
3	NET101	Networking Essentials	IP addressing and basic network setup.	Edit Delete
1	WEB101	Web Development Fundamentals	HTML, CSS and basic PHP.	Edit Delete

Figure 2b. Course list after saving, showing the new course and the success message.

Screenshot 3 — Class Management

How to capture:

1. Open admin/classes.php
2. Capture the Class List showing course, schedule and Slots Left. Note that WEB101-A has 5 slots.

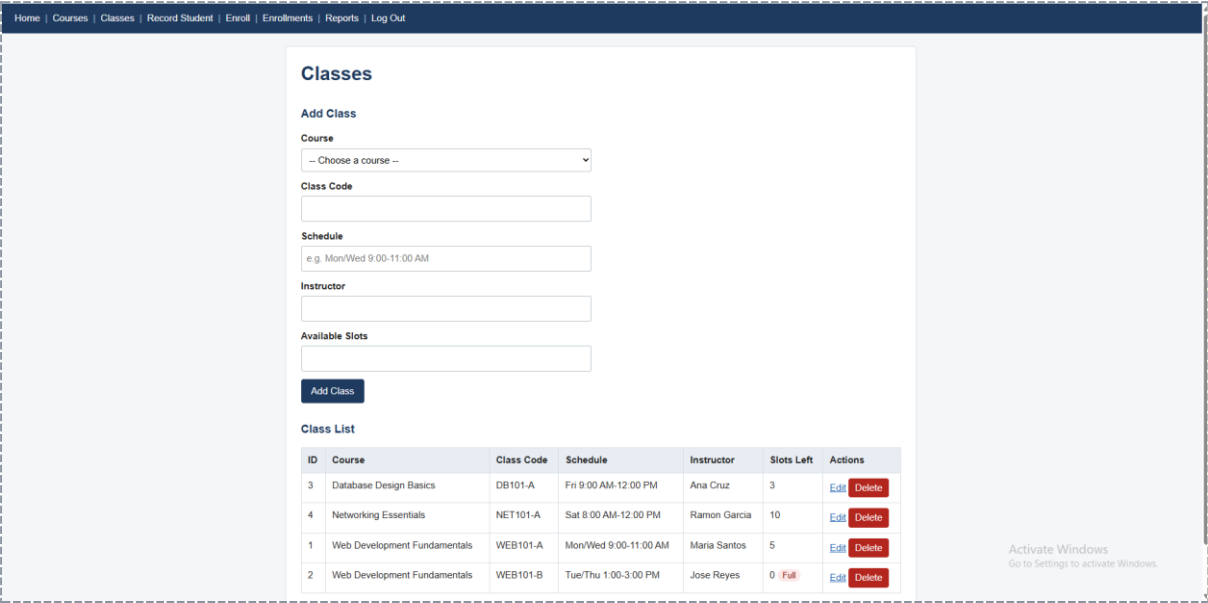


Figure 3. Classes page showing each class with its course, schedule and remaining slots.

Screenshot 4 — Student Recording Form

How to capture:

1. Open admin/students.php
2. Fill in a name, email and phone, and choose class WEB101-A (5 slots left).
3. Capture the filled form before clicking Record Student.

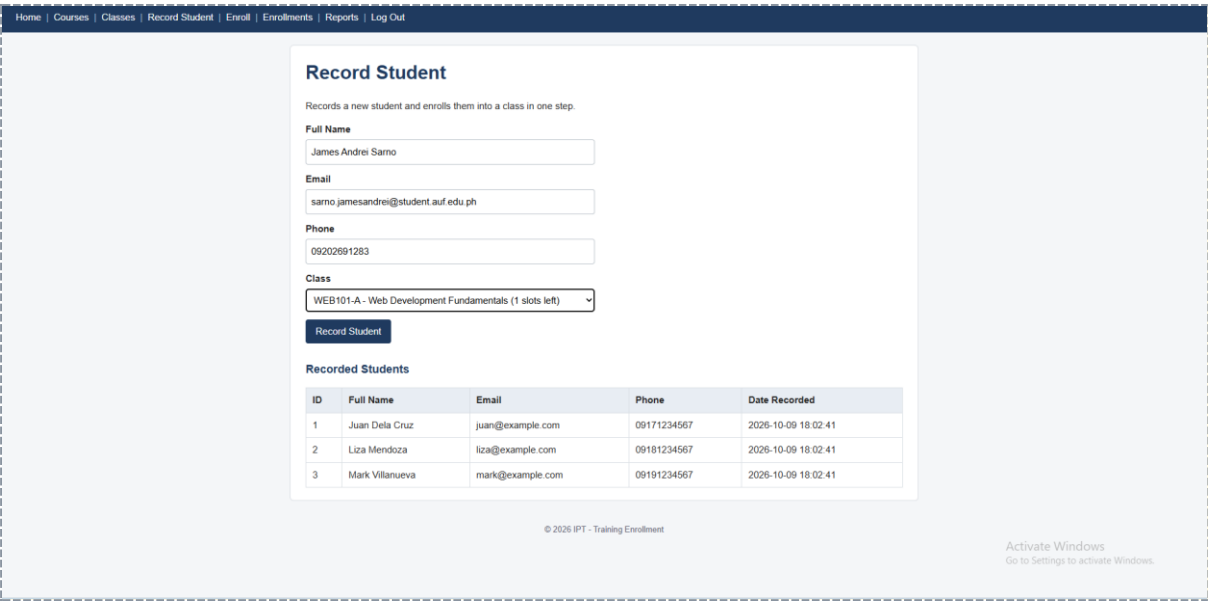


Figure 4. Student recording form with name, email, phone and the selected class.

Screenshot 5 — Successful Recording

How to capture:

1. Click Record Student. Capture the green message and the new row in the Recorded Students table (5a).
2. Open Enrollments. Capture the new enrollment row at the top (5b).

Home | Courses | Classes | Record Student | Enroll | Enrollments | Reports | Log Out

Student recorded and enrolled. (Student ID 4)

Record Student

Records a new student and enrolls them into a class in one step.

Full Name

Email

Phone

Class
 -- Choose a class --

Record Student

Recorded Students

ID	Full Name	Email	Phone	Date Recorded
4	James Andrei Sarno	sarno.jamesandrei@student.auf.edu.ph	09202691283	2026-10-09 18:22:46
1	Juan Dela Cruz	juan@example.com	09171234567	2026-10-09 18:02:41
2	Liza Mendoza	liza@example.com	09181234567	2026-10-09 18:02:41
3	Mark Villanueva	mark@example.com	09191234567	2026-10-09 18:02:41

Activate Windows
Go to Settings to activate Windows.

Figure 5a. Confirmation message and the new row in the students list.

Home | Courses | Classes | Record Student | Enroll | Enrollments | Reports | Log Out

Enrollments

Search student, class or course All statuses

4 enrollment(s) found.

ID	Student	Course	Class	Schedule	Instructor	Date Enrolled	Status	Action
4	James Andrei Sarno	Web Development Fundamentals	WEB101-A	Wed 9:00-11:00 AM	Maria Santos	2026-10-09 18:22:46	active	Cancel
3	Mark Villanueva	Database Design Basics	DB101-A	Fri 9:00 AM-12:00 PM	Ana Cruz	2026-10-09 18:02:41	active	Cancel
2	Liza Mendoza	Web Development Fundamentals	WEB101-B	Tue/Thu 1:00-3:00 PM	Jose Reyes	2026-10-09 18:02:41	active	Cancel
1	Juan Dela Cruz	Web Development Fundamentals	WEB101-B	Tue/Thu 1:00-3:00 PM	Jose Reyes	2026-10-09 18:02:41	active	Cancel

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Figure 5b. The new row in the enrollments list.

Screenshot 6 — Enrollments List

How to capture:

1. Open admin/enrollments.php
2. Capture the table showing student, course, class, schedule, instructor and status.

Home | Courses | Classes | Record Student | Enroll | Enrollments | Reports | Log Out

Enrollments

Search student, class or course: All statuses [Reset](#)

4 enrollment(s) found.

ID	Student	Course	Class	Schedule	Instructor	Date Enrolled	Status	Action
4	James Andrei Sarno	Web Development Fundamentals	WEB101-A	Wed 9:00-11:00 AM	Maria Santos	2026-10-09 18:22:46	active	Cancel
3	Mark Villanueva	Database Design Basics	DB101-A	Fri 9:00 AM-12:00 PM	Ana Cruz	2026-10-09 18:02:41	active	Cancel
2	Liza Mendoza	Web Development Fundamentals	WEB101-B	Tue/Thu 1:00-3:00 PM	Jose Reyes	2026-10-09 18:02:41	active	Cancel
1	Juan Dela Cruz	Web Development Fundamentals	WEB101-B	Tue/Thu 1:00-3:00 PM	Jose Reyes	2026-10-09 18:02:41	active	Cancel

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Figure 6. Enrollments list joined with student, class and course details.

Screenshot 7 — Cancelled Enrollment

How to capture:

1. On the Enrollments page, click Cancel on the enrollment you made in Screenshot 5 and confirm.
2. Capture the message and the row now showing status "cancelled" (7a).
3. Open Classes. Capture WEB101-A back to 5 slots left (7b).

Home | Courses | Classes | Record Student | Enroll | Enrollments | Reports | Log Out

Enrollments

Search student, class or course: All statuses [Reset](#)

4 enrollment(s) found.

ID	Student	Course	Class	Schedule	Instructor	Date Enrolled	Status	Action
4	James Andrei Sarno	Web Development Fundamentals	WEB101-A	Wed 9:00-11:00 AM	Maria Santos	2026-10-09 18:22:46	cancelled	
3	Mark Villanueva	Database Design Basics	DB101-A	Fri 9:00 AM-12:00 PM	Ana Cruz	2026-10-09 18:02:41	active	Cancel
2	Liza Mendoza	Web Development Fundamentals	WEB101-B	Tue/Thu 1:00-3:00 PM	Jose Reyes	2026-10-09 18:02:41	active	Cancel
1	Juan Dela Cruz	Web Development Fundamentals	WEB101-B	Tue/Thu 1:00-3:00 PM	Jose Reyes	2026-10-09 18:02:41	active	Cancel

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Figure 7a. Enrollment with status cancelled.

Classes

Add Class

Course: -- Choose a course --

Class Code:

Schedule: e.g. Mon/Wed 9:00-11:00 AM

Instructor:

Available Slots:

Add Class

Class List

ID	Course	Class Code	Schedule	Instructor	Slots Left	Actions
3	Database Design Basics	DB101-A	Fri 9:00 AM-12:00 PM	Ana Cruz	3	Edit Delete
4	Networking Essentials	NET101-A	Sat 8:00 AM-12:00 PM	Ramon Garcia	10	Edit Delete
1	Web Development Fundamentals	WEB101-A	Mon/Wed 9:00-11:00 AM	Maria Santos	5	Edit Delete
5	Web Development Fundamentals	WEB101-A	Wed 9:00-11:00 AM	Maria Santos	1	Edit Delete

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Figure 7b. Class WEB101-A with its slot restored (4 became 5).

Screenshot 8 — Rollback Case

How to capture:

1. Open Record Student. Type a new name (for example "Rollback Test") and choose class WEB101-B (FULL).
2. Click Record Student. Capture the red "No slots available. Nothing was saved." message together with the Recorded Students table below it. The new name must not be in the table (8a).
3. Open Enrollments. Capture the list showing that no new enrollment was added (8b).

Record Student

Records a new student and enrolls them into a class in one step.

No slots available. Nothing was saved.

Full Name: Christian Madiambayan

Email: madiambayan.christiananthony@student.auf.edu.ph

Phone: 09397127438

Class: WEB101-B - Web Development Fundamentals (FULL)

Record Student

Recorded Students

ID	Full Name	Email	Phone	Date Recorded
4	James Andrei Sarno	sarno.jamesandrei@student.auf.edu.ph	09202691283	2026-10-09 18:22:46
1	Juan Dela Cruz	juan@example.com	09171234567	2026-10-09 18:02:41
2	Liza Mendoza	liza@example.com	09181234567	2026-10-09 18:02:41
3	Mark Villanueva	mark@example.com	09191234567	2026-10-09 18:02:41

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Figure 8a. "No slots available" message when recording into a full class. The student was not saved.

Home Courses Classes Record Student Enroll Enrollments Reports Log Out																																																					
Enrollments Search student, class or course All statuses Search Reset 4 enrollment(s) found. <table> <tr> <th>ID</th><th>Student</th><th>Course</th><th>Class</th><th>Schedule</th><th>Instructor</th><th>Date Enrolled</th><th>Status</th><th>Action</th></tr> <tr> <td>4</td><td>James Andrei Sarno</td><td>Web Development Fundamentals</td><td>WEB101-A</td><td>Wed 9:00-11:00 AM</td><td>Maria Santos</td><td>2026-10-09 18:22:46</td><td>cancelled</td><td></td></tr> <tr> <td>3</td><td>Mark Villanueva</td><td>Database Design Basics</td><td>DB101-A</td><td>Fri 9:00 AM-12:00 PM</td><td>Ana Cruz</td><td>2026-10-09 18:02:41</td><td>active</td><td>Cancel</td></tr> <tr> <td>2</td><td>Liza Mendoza</td><td>Web Development Fundamentals</td><td>WEB101-B</td><td>Tue/Thu 1:00-3:00 PM</td><td>Jose Reyes</td><td>2026-10-09 18:02:41</td><td>active</td><td>Cancel</td></tr> <tr> <td>1</td><td>Juan Dela Cruz</td><td>Web Development Fundamentals</td><td>WEB101-B</td><td>Tue/Thu 1:00-3:00 PM</td><td>Jose Reyes</td><td>2026-10-09 18:02:41</td><td>active</td><td>Cancel</td></tr> </table> © 2026 IPT - Training Enrollment									ID	Student	Course	Class	Schedule	Instructor	Date Enrolled	Status	Action	4	James Andrei Sarno	Web Development Fundamentals	WEB101-A	Wed 9:00-11:00 AM	Maria Santos	2026-10-09 18:22:46	cancelled		3	Mark Villanueva	Database Design Basics	DB101-A	Fri 9:00 AM-12:00 PM	Ana Cruz	2026-10-09 18:02:41	active	Cancel	2	Liza Mendoza	Web Development Fundamentals	WEB101-B	Tue/Thu 1:00-3:00 PM	Jose Reyes	2026-10-09 18:02:41	active	Cancel	1	Juan Dela Cruz	Web Development Fundamentals	WEB101-B	Tue/Thu 1:00-3:00 PM	Jose Reyes	2026-10-09 18:02:41	active	Cancel
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Activate Windows Go to Settings to activate Windows.																																																					

Figure 8b. Enrollments list with no new row, proving the transaction was rolled back.

Screenshot 9 — Database Evidence

How to capture:

1. Open <http://localhost/phpmyadmin> and click the training_db database.
2. Click the students table and capture the rows (9a).
3. Click the enrollments table and capture the rows (9b).
4. Click the classes table and capture the rows (9c).

phpMyAdmin Recent Favourites - New - expense_db - finals - information_schema - ip10_lab - mysql - performance_schema - portfolio_db - restaurant - spendify_db - test - training_db - New - classes - courses - enrollments students																																			
Server: 127.0.0.1 • Database: training_db • Table: students Showing rows 0 - 3 (4 total, Query took 0.0002 seconds) SELECT * FROM "students" ☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh] ☐ Show all Number of rows: 25 Filter rows: Search this table Sort by key: None Extra options <table> <tr> <th></th><th>student_id</th><th>full_name</th><th>email</th><th>phone</th><th>created_at</th></tr> <tr> <td>☐ Edit Copy Delete</td><td>1</td><td>Juan Dela Cruz</td><td>juan@example.com</td><td>09171234567</td><td>2026-10-09 18:02:41</td></tr> <tr> <td>☐ Edit Copy Delete</td><td>2</td><td>Liza Mendoza</td><td>liza@example.com</td><td>09181234567</td><td>2026-10-09 18:02:41</td></tr> <tr> <td>☐ Edit Copy Delete</td><td>3</td><td>Mark Villanueva</td><td>mark@example.com</td><td>09191234567</td><td>2026-10-09 18:02:41</td></tr> <tr> <td>☐ Edit Copy Delete</td><td>4</td><td>James Andrei Sarno</td><td>sarno.jamesandrei@student.auf.edu.ph</td><td>09202691283</td><td>2026-10-09 18:22:46</td></tr> </table> ☐ Check all With selected: Edit Copy Delete Export ☐ Show all Number of rows: 25 Filter rows: Search this table Sort by key: None Query results operations Print Copy to clipboard Export Display chart Create view Bookmark this SQL query Label: ☐ Let every user access this bookmark Bookmark this SQL query							student_id	full_name	email	phone	created_at	☐ Edit Copy Delete	1	Juan Dela Cruz	juan@example.com	09171234567	2026-10-09 18:02:41	☐ Edit Copy Delete	2	Liza Mendoza	liza@example.com	09181234567	2026-10-09 18:02:41	☐ Edit Copy Delete	3	Mark Villanueva	mark@example.com	09191234567	2026-10-09 18:02:41	☐ Edit Copy Delete	4	James Andrei Sarno	sarno.jamesandrei@student.auf.edu.ph	09202691283	2026-10-09 18:22:46
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localhost/phpmyadmin/index.php Console Activate Windows Go to Settings to activate Windows.																																			

Figure 9a. phpMyAdmin view of the students table.

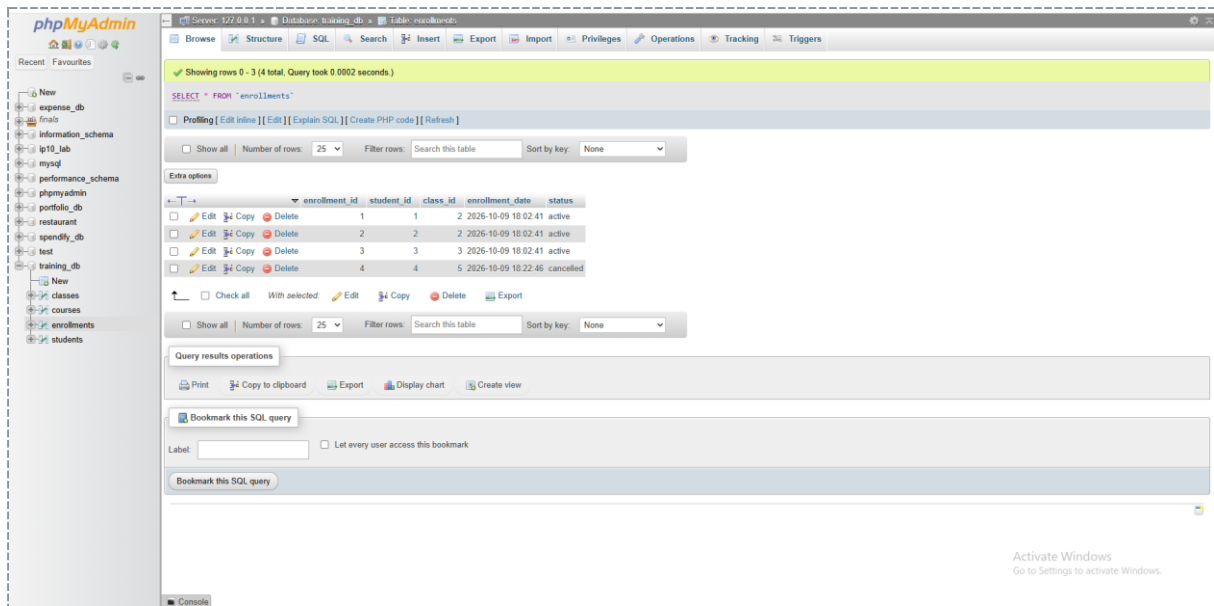


Figure 9b. phpMyAdmin view of the enrollments table.

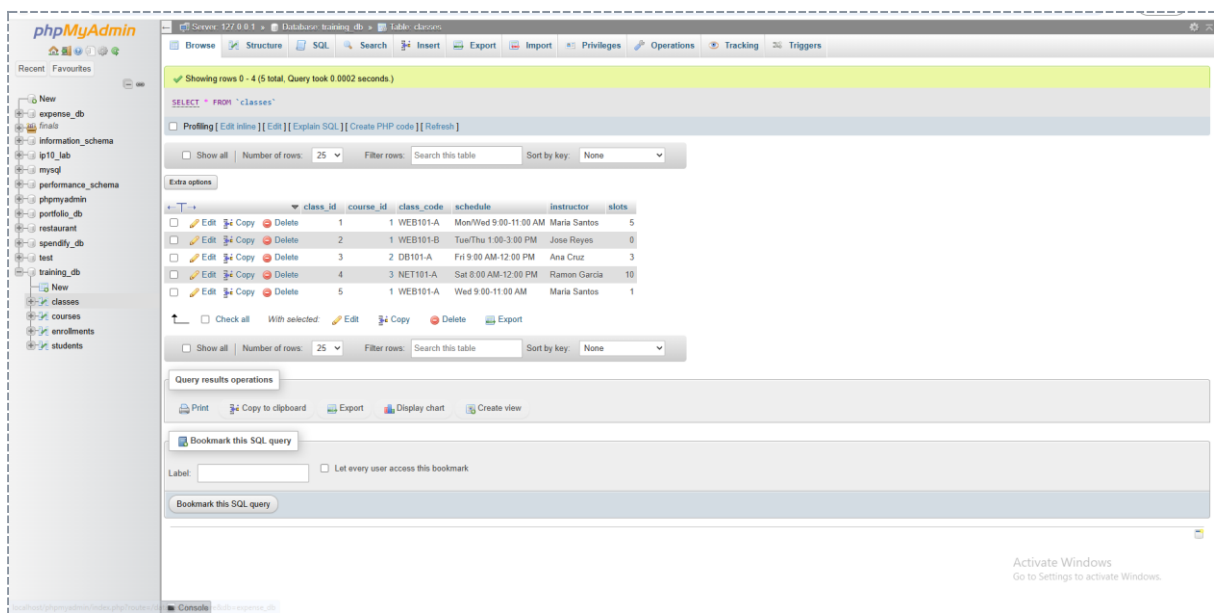


Figure 9c. phpMyAdmin view of the classes table.